

Mohamed Issaoui

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Profile

Final-year **Computer Science Engineering Student (AI Track)** at ENSI (national competitive exam rank 56/650), graduating in **September 2026**. Interested in **deep learning, reinforcement learning, embodied AI, and vision-language navigation for autonomous robots**.

Education

Engineering Preparatory Institute of Monastir – National Competitive Exam Rank: 56/650 2021–2023
National School of Computer Science (ENSI), Tunisia – Engineering Degree (AI Track) 2023–2026

Experience

AI/Robotics Engineer Intern – OORB, Tunisia Jun 2025 – Aug 2025

- Designed **quadruped locomotion and manipulation RL environments** with explicit state and reward representations in MuJoCo/Gymnasium.
- Trained and deployed **PPO-based policies** integrated into ROS2 nodes for real-time control.
- Built reproducible and scalable experimentation pipelines using **Docker, Kubeflow, and TensorBoard** on **Linux and ROS2**.

Research Intern (Embodied AI / VLN) – SPIN Lab, Wilfrid Laurier University, Canada Apr 2026 – Jul 2026

- Designed a **modular Vision-Language Navigation (VLN)** research prototype targeting later deployment on a real **ROS2** robot.
- Built a **predictive world model** maintaining probabilistic beliefs over unseen spaces, combining **semantic priors, hypothesis generation, and belief updating** to guide navigation.
- Integrated **VLM-based perception** producing a grounded **scene-graph memory**, used by an **LLM planner** for subgoal generation and a stateful navigation policy with frontier scoring.
- Experimented with a **generative anticipation module** to imagine likely future observations under high uncertainty, feeding hypotheses back into the belief state. *Co-authoring a research paper on vision-language navigation (in preparation).*

Projects

Autonomous Mobile Robot – Eurobot Competition 2024

- Implemented **YOLO-based perception** and LiDAR-based SLAM for environment understanding.
- Designed **PID control pipelines** using encoder feedback for reliable navigation.
- Deployed complete robotic stacks on **Raspberry Pi and Jetson Nano** using ROS2.

LLM-Guided Robot Learning – ENSI & OORB 2025

🌐 [GitHub Repo](#)

- Developed an **LLM-guided reinforcement learning framework** inspired by NVIDIA Eureka.
- Used language models to **iteratively reason over environment feedback** and refine reward functions.
- Evaluated reward efficiency and behavior quality on MuJoCo robotic tasks with sparse rewards.

Technical Skills

Languages: Python, C++, C **AI:** PyTorch, CNNs, Transformers, Vision-Language Models, LLMs, World Models

Robot Learning: Reinforcement Learning, imitation learning, Diffusion Policies

Robotics: ROS2, Gazebo, MuJoCo, SLAM, Control Systems, embodied simulation, Isaac Sim/Lab

Languages

Arabic (Native) French (Fluent) English (Fluent)

Certifications

Machine Learning Specialization – Coursera

NVIDIA Jetson Nano AI Development